

Data Types in Python

1. String :

- A **String** is a sequence of characters written inside single quotes (' ') or double quotes (" ") or Triple quotes (''' ''') .

➤ String Characteristics / Properties

Property	Description
Ordered	Characters follow fixed order
Immutable	Cannot change after creation
Indexed	Access characters using index
Slicing Supported	Can access part of string
Iterable	Can use loop on string
Allows Duplicates	Same characters allowed
Sequence Type	Stores sequence of characters
Homogeneous	Stores only characters
Supports Positive Indexing	Index starts from 0
Supports Negative Indexing	Reverse indexing starts from -1
Concatenation Supported	Can join strings using +
Repetition Supported	Can repeat
Dynamic Memory Allocation	Size managed automatically

➤ String Methods

Method	Description
upper()	Converts string to uppercase
lower()	Converts string to lowercase
capitalize()	Capitalizes first character
title()	Capitalizes first letter of every word
swapcase()	Changes uppercase to lowercase and vice versa
strip()	Removes spaces from both sides
lstrip()	Removes left-side spaces
rstrip()	Removes right-side spaces
replace(old, new)	Replaces text
split()	Splits string into list
join()	Joins elements into string
find()	Finds position of character/word
index()	Returns index position
count()	Counts occurrences
startswith()	Checks starting text
endswith()	Checks ending text
isalpha()	Checks alphabet only
isdigit()	Checks digits only
isalnum()	Checks alphabets and numbers

➤ Built-in Functions Used with String

Function	Description
len()	Returns length of string
max()	Returns largest character
min()	Returns smallest character
sorted()	Returns sorted characters
type()	Returns datatype
list()	Converts string to list
tuple()	Converts string to tuple
set()	Converts string to set
str()	Converts value to string
enumerate()	Returns index with value
reversed()	Reverses string iterator
ord()	Returns ASCII value
chr()	Returns character from ASCII
any()	Returns True if any character is True
all()	Returns True if all characters are True

2. List :

- **List** is a collection data type used to store multiple values in a single variable.
- Lists are written inside square brackets []

➤ List Characteristics / Properties

Property	Description
Ordered	Elements follow fixed order
Mutable	Can change after creation
Indexed	Access elements using index
Slicing Supported	Can access part of list
Iterable	Can use loop on list
Allows Duplicates	Duplicate values allowed
Dynamic	Size can increase or decrease
Heterogeneous	Can store different datatypes
Homogeneous	Can also store same datatype
Nested Supported	List inside another list possible
Supports Positive Indexing	Index starts from 0
Supports Negative Indexing	Reverse indexing starts from -1
Concatenation Supported	Can join lists using +
Repetition Supported	Can repeat
Dynamic Memory Allocation	Memory managed automatically

➤ List Methods

Method	Description
append()	Adds element at end
extend()	Adds multiple elements
insert()	Inserts element at specific position
remove()	Removes specified element
pop()	Removes element using index
clear()	Removes all elements
index()	Returns index position
count()	Counts occurrences
sort()	Sorts list
reverse()	Reverses list
copy()	Creates copy of list

➤ Built-in Functions Used with List

Function	Description
len()	Returns total elements
max()	Returns largest element
min()	Returns smallest element
sum()	Returns total sum
sorted()	Returns sorted list
type()	Returns datatype
list()	Converts data into list
tuple()	Converts list to tuple
set()	Converts list to set
str()	Converts list to string
enumerate()	Returns index with value
reversed()	Returns reverse iterator
any()	Returns True if any element is True
all()	Returns True if all elements are True

3. Tuple :

- A **Tuple** is a collection data type used to store multiple values in a single variable.
- Tuples are written inside parentheses ().

➤ Tuple Characteristics / Properties

Property	Description
Ordered	Elements follow fixed order
Immutable	Cannot change after creation
Indexed	Access elements using index
Slicing Supported	Can access part of tuple
Iterable	Can use loop on tuple
Allows Duplicates	Duplicate values allowed
Heterogeneous	Can store different datatypes
Homogeneous	Can also store same datatype
Nested Supported	Tuple inside another tuple possible
Supports Positive Indexing	Index starts from 0
Supports Negative Indexing	Reverse indexing starts from -1
Concatenation Supported	Can join tuples using +
Repetition Supported	Can repeat

➤ Tuple Methods

Method	Description
count()	Counts occurrences of element
index()	Returns index position

➤ Built-in Functions Used with Tuple

Function	Description
len()	Returns total elements
max()	Returns largest element
min()	Returns smallest element
sum()	Returns total sum
sorted()	Returns sorted list from tuple
type()	Returns datatype
tuple()	Converts data into tuple
list()	Converts tuple to list
set()	Converts tuple to set
str()	Converts tuple to string
enumerate()	Returns index with value
reversed()	Returns reverse iterator
any()	Returns True if any element is True
all()	Returns True if all elements are True

4. Set :

- A **Set** is a collection data type used to store multiple unique values in a single variable.
- Sets are written inside curly brackets { }.

➤ Set Characteristics / Properties

Property	Description
Unordered	Elements do not follow fixed order
Mutable	Can change after creation
Unindexed	Cannot access using index
No Duplicates Allowed	Duplicate values automatically removed
Iterable	Can use loop on set
Heterogeneous	Can store different datatypes
Homogeneous	Can also store same datatype
Dynamic	Size can increase or decrease
Nested Set Not Allowed	Normal sets cannot store sets
Supports Mathematical Operations	Union, intersection, difference etc.
Fast Searching	Searching is faster compared to list
Uses Hashing Technique	Internal hashing used for storage
Mutable but Elements Immutable	Elements should be immutable
No Slicing	Slicing not supported
No Positive/Negative Indexing	Indexing not supported

➤ Set Methods

Method	Description
add()	Adds one element
update()	Adds multiple elements
remove()	Removes element (error if not found)
discard()	Removes element safely
pop()	Removes random element
clear()	Removes all elements
copy()	Creates copy of set
union()	Combines sets

➤ Built-in Functions Used with Set

Function	Description
len()	Returns total elements
max()	Returns largest element
min()	Returns smallest element
sum()	Returns total sum
sorted()	Returns sorted list from set
type()	Returns datatype
set()	Converts data into set

Function	Description
list()	Converts set to list
tuple()	Converts set to tuple
str()	Converts set to string
enumerate()	Returns index with value
any()	Returns True if any element is True
all()	Returns True if all elements are True

5. Dictionary :

- A **Dictionary** is a collection data type used to store data in **key : value** pairs.
- Dictionaries are written inside curly brackets { }.

➤ Dictionary Characteristics / Properties

Property	Description
Mutable	Can change after creation
Ordered	Maintains insertion order
Key-Value Pair	Data stored in key:value format
Keys Must Be Unique	Duplicate keys not allowed
Values Can Be Duplicate	Same values allowed
Indexed by Keys	Access values using keys
Iterable	Can use loop on dictionary

Property	Description
Heterogeneous	Can store different datatypes
Homogeneous	Can also store same datatype
Dynamic	Size can increase or decrease
Nested Dictionary Supported	Dictionary inside dictionary possible
No Numeric Indexing	Cannot access using numeric index
No Slicing	Slicing not supported
Fast Searching	Searching using keys is fast
Uses Hashing Technique	Internal hashing used for keys
Keys Must Be Immutable	Keys should be immutable datatype

➤ Dictionary Methods

Method	Description
keys()	Returns all keys
values()	Returns all values
items()	Returns key-value pairs
get()	Returns value using key
update()	Updates dictionary
pop()	Removes specified key
popitem()	Removes last inserted item
clear()	Removes all items
copy()	Creates copy of dictionary

Method	Description
dict()	Creates dictionary
count()	Not supported in dictionary
index()	Not supported in dictionary

➤ Built-in Functions Used with Dictionary

Function	Description
len()	Returns total key-value pairs
max()	Returns largest key
min()	Returns smallest key
sorted()	Returns sorted keys
type()	Returns datatype
dict()	Converts data into dictionary
list()	Converts dictionary to list of keys
tuple()	Converts dictionary to tuple of keys
set()	Converts dictionary to set of keys
str()	Converts dictionary to string
enumerate()	Returns index with keys
any()	Returns True if any key is True
all()	Returns True if all keys are True